

Chemical Accuracy on NISQ Devices: A Comparative Study of VQE Ansätze and Error Mitigation

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Abstract

TODO: to be written once Phase 7 results exist. Placeholder scaffold only – no results are claimed yet.

1 Introduction

TODO (Phase 1+): motivate the research question – under what conditions is chemical accuracy ($|\Delta E| < 1.6$ mHa) achievable on NISQ devices via VQE, and what limits it.

2 Theory

TODO (Phase 1–2): Born-Oppenheimer approximation, second quantization, fermion-to-qubit mappings, the VQE variational principle.

3 Methods

TODO (Phase 3–6): ansatz library, optimizers, noise models, error mitigation techniques, hardware access.

4 Results

TODO (Phase 7): bond dissociation curves, convergence statistics, error mitigation waterfall analysis, error budget decomposition.

5 Discussion

TODO (Phase 7).

6 Limitations

TODO – to be taken seriously, not a formality.